

Setting. At window w the state carried from earlier windows is the sketch B and the row sample (reservoir) R . Let

$$N_w := \begin{pmatrix} A_1 \\ \vdots \\ A_w \end{pmatrix}, \quad C_w := \begin{pmatrix} R \\ A_w \end{pmatrix}, \quad (1)$$

and, for a nonzero y ,

$$h_2(y) := -\log\left(\frac{\|y\|_4^4}{\|y\|_2^4}\right) = 4\log\|y\|_2 - 4\log\|y\|_4. \quad (2)$$

Throughout $v \in \mathbb{S}^{d-1}$ and $A \in \mathbb{R}^{n \times d}$; assume $\text{row}(C_w) > 1$ and $\text{row}(N_w) \leq n$.

Assumptions. Assume

$$\|Bv\|_2^2 + \|A_w v\|_2^2 \approx \|N_w v\|_2^2, \quad (A1)$$

$$\frac{h_2(C_w v)}{\log \text{row}(C_w)} \approx \frac{h_2(N_w v)}{\log \text{row}(N_w)} \approx \frac{h_2(Av)}{\log n}. \quad (A2)$$

Resulting inequality. Under (A2) and $\|N_w v\|_4 \leq \|Av\|_4$,

$$\|Av\|_2^2 \gtrsim \|N_w v\|_2^2 \left(\frac{\text{row}(N_w)}{n} \right)^{-h_2(N_w v)/(2 \log \text{row}(N_w))}. \quad (\star)$$

The combined score. Substituting (A1) and (A2) into $(\star)$ gives

$$\|Av\|_2^2 \gtrsim (\|Bv\|_2^2 + \|A_w v\|_2^2) \left(\frac{\text{row}(N_w)}{n} \right)^{-h_2(C_w v)/(2 \log \text{row}(C_w))} =: \text{score}_w(v), \quad (3)$$

with

$$h_2(C_w v) = -\log\left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2}\right). \quad (4)$$

If no old row sample is available, take $R = \emptyset$. If no old sketch is available, take $B = 0$.

Algorithm overview. Process one row-window at a time.

For the first window, there is no previous sketch and no old row sample, so

$$B = 0, \quad R = \emptyset, \quad (5)$$

and $C_1 = A_1$, so

$$\text{score}_1(v) = \|A_1 v\|_2^2 \left(\frac{\text{row}(A_1)}{n} \right)^{-\frac{h_2(A_1 v)}{2 \log \text{row}(A_1)}}. \quad (6)$$

Compute retained directions $v_1, \dots, v_r$ sequentially by solving

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j} \text{score}_1(v). \quad (7)$$

Collecting them gives a retained right basis

$$V_{\text{hat}} = [v_1, \dots, v_r]. \quad (8)$$

With the current carried operator $M = A_1$, form

$$Y = MV_{\text{hat}}, \quad U = \text{orth}(Y), \quad (9)$$

compute the SVD of

$$U^\top M = U_{\text{small}} \Sigma V_{\text{new}}^\top, \quad (10)$$

and carry the sketch

$$B_{\text{next}} = \Sigma V_{\text{new}}^\top. \quad (11)$$

Also update the row sample R_{next} from the rows seen so far.

For a later window w , the input state is the carried sketch $B^{(w-1)} = \Sigma_{w-1} V_{w-1}^\top$ and the row sample $R^{(w-1)}$. Optimize the combined objective $\text{score}_w(v)$ above and compute orthonormal retained directions sequentially by solving

$$v_j^{(w)} \in \arg \max_{v \in \mathbb{S}^{d-1}, (v_i^{(w)})^\top v = 0 \ \forall i < j} \text{score}_w(v). \quad (12)$$

After selecting the new right space, convert that space into the left-projected carry sketch. With

$$\hat{V}_w = [v_1^{(w)}, \dots, v_r^{(w)}], \quad M_w := \begin{pmatrix} B^{(w-1)} \\ A_w \end{pmatrix}, \quad (13)$$

form

$$Y_w := M_w \hat{V}_w, \quad U_{,w} := \text{orth}(Y_w), \quad (14)$$

and compute

$$U_{,w}^\top M_w = U_{\text{small},w} \Sigma_w V_w^\top. \quad (15)$$

The sketch passed to the next window is

$$B^{(w)} = \Sigma_w V_w^\top. \quad (16)$$

The row sample is updated so that $R^{(w)}$ is drawn from the raw rows seen through window w , not from the sketch rows.

Why this is the sketch carried into the next window. Consider the naive alternative of carrying V together with the column norms $s_i := \|M_w v_i\|_2$. With

$$M_w = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad V = [e_1, e_2], \quad (17)$$

one has $s_1 = s_2 = 1$, so $B_{\text{diag}} := \text{diag}(s)V^\top = I$ gives $\|B_{\text{diag}}v\|_2^2 = 1$ uniformly on $\mathbb{S}^1$, whereas

$$\|M_w v\|_2^2 = v^\top \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v \quad (18)$$

vanishes at $v = (1, -1)/\sqrt{2}$ (outputs cancel) and equals 2 at $v = (1, 1)/\sqrt{2}$ (outputs reinforce). Column norms drop the cross correlations $(M_w v_i)^\top (M_w v_j)$, which are generically nonzero because the combined score pulls V away from the right singular directions of M_w . The window therefore receives

$$B_{\text{next}} := \Sigma V_{\text{new}}^\top, \quad U^\top M_w = U_{\text{small}} \Sigma V_{\text{new}}^\top, \quad (19)$$

the norm-preserving right-action form of $P_U M_w = U U^\top M_w$, where $P_U := U U^\top$ is the orthogonal projector onto the retained left space. For any v , $\|B_{\text{next}} v\|_2 = \|P_U M_w v\|_2$, and for $v \in \text{span}(V)$ this equals $\|M_w v\|_2$ exactly since $M_w v \in \text{span}(U)$. Hence $B_{\text{next}}^\top B_{\text{next}}$ reproduces the full Gram $V^\top M_w^\top M_w V$ on the retained right span, all off-diagonals included. The row sample R_{next} enters the pooled entropy in the next window.

Combined entropy. With $y := C_w v$, the definition of h_2 gives

$$h_2(C_w v) = 4 \log \|y\|_2 - 4 \log \|y\|_4. \quad (20)$$

This is the same formula as in the single-block case, applied to the stacked vector combining the old row sample Rv with the current window $A_w v$. The entropy is computed from the pooled vector, so rare large entries in either Rv or $A_w v$ affect the same correction factor.

Simplified form. Let

$$c_w := \frac{\log(\text{row}(N_w)/n)}{2 \log \text{row}(C_w)}, \quad y := C_w v. \quad (21)$$

Since

$$h_2(C_w v) = -\log \left(\frac{\|y\|_4^4}{\|y\|_2^4} \right), \quad (22)$$

the entropy multiplier is

$$\left(\frac{\text{row}(N_w)}{n} \right)^{-h_2(C_w v)/(2 \log \text{row}(C_w))} = \left(\frac{\|y\|_4^4}{\|y\|_2^4} \right)^{c_w} = \|C_w v\|_4^{4c_w} \|C_w v\|_2^{-4c_w}. \quad (23)$$

Thus

$$\text{score}_w(v) = (\|Bv\|_2^2 + \|A_w v\|_2^2) \phi_w(v), \quad (24)$$

where

$$\phi_w(v) := \|C_w v\|_4^{4c_w} \|C_w v\|_2^{-4c_w}. \quad (25)$$

When $\text{row}(N_w) < n$, $c_w < 0$, and high-entropy pooled vectors are boosted by the combined correction.

Exact gradient. Let

$$y_B := Bv, \quad z := A_w v, \quad y := C_w v, \quad (26)$$

and assume the denominators below are nonzero. The gradient of $\|Bv\|_2^2 + \|A_w v\|_2^2$ is

$$\nabla (\|Bv\|_2^2 + \|A_w v\|_2^2) = 2B^\top Bv + 2A_w^\top A_w v. \quad (27)$$

For the entropy multiplier,

$$\log \phi_w(v) = 4c_w \log \|C_w v\|_4 - 4c_w \log \|C_w v\|_2. \quad (28)$$

Differentiating gives

$$\nabla \log \phi_w(v) = 4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2}. \quad (29)$$

Hence

$$\nabla \phi_w(v) = \phi_w(v) \left[4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2} \right]. \quad (30)$$

The full Euclidean gradient is therefore

$$\nabla \text{score}_w(v) = \phi_w(v) (2B^\top Bv + 2A_w^\top A_w v) \quad (31)$$

$$+ (\|Bv\|_2^2 + \|A_w v\|_2^2) \phi_w(v) \left[4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2} \right]. \quad (32)$$

Equivalently,

$$\nabla \text{score}_w(v) = \text{score}_w(v) \left[\frac{2B^\top Bv + 2A_w^\top A_w v}{\|Bv\|_2^2 + \|A_w v\|_2^2} + 4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2} \right], \quad (33)$$

provided $\|Bv\|_2^2 + \|A_w v\|_2^2 > 0$.

Expanded row-sample form. Because

$$C_w^\top C_w v = R^\top R v + A_w^\top A_w v, \quad (34)$$

and

$$C_w^\top (C_w v)^{\odot 3} = R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}, \quad (35)$$

the entropy-gradient part can be evaluated without explicitly forming C_w :

$$C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) = \frac{R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}}{\|R v\|_4^4 + \|A_w v\|_4^4}, \quad (36)$$

and

$$\frac{C_w^\top C_w v}{\|C_w v\|_2^2} = \frac{R^\top R v + A_w^\top A_w v}{\|R v\|_2^2 + \|A_w v\|_2^2}. \quad (37)$$

Likewise,

$$\phi_w(v) = \left(\frac{\|R v\|_4^4 + \|A_w v\|_4^4}{(\|R v\|_2^2 + \|A_w v\|_2^2)^2} \right)^{c_w}. \quad (38)$$

Stored-basis form. If

$$B = S V^\top, \quad V = [v_1, \dots, v_r], \quad S = \text{diag}(s_1, \dots, s_r), \quad (39)$$

then

$$\|B v\|_2^2 = \sum_{j=1}^r s_j^2 (v_j^\top v)^2, \quad B^\top B v = \sum_{j=1}^r s_j^2 (v_j^\top v) v_j. \quad (40)$$

Thus the energy-gradient part of the combined score is

$$2B^\top B v + 2A_w^\top A_w v = 2 \sum_{j=1}^r s_j^2 (v_j^\top v) v_j + 2A_w^\top A_w v. \quad (41)$$

Optimization problem. To extract a direction, solve

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v = 0} \text{score}_w(v), \quad (42)$$

where the columns of Q are the previously accepted directions.

Projected full-gradient ascent. Use the exact gradient of the combined score. Project it onto the feasible tangent space:

$$g_{\text{tan}}(v) := (I - Q Q^\top - v v^\top) \nabla \text{score}_w(v). \quad (43)$$

Then take

$$\tilde{v}^+ = v + \eta g_{\text{tan}}(v), \quad v^+ = \frac{(I - Q Q^\top) \tilde{v}^+}{\|(I - Q Q^\top) \tilde{v}^+\|_2}, \quad (44)$$

with $\eta > 0$ chosen by backtracking line search on the true combined score.

Basis-expansion interpretation. If one is building a local search subspace rather than taking a pure steepest-ascent step, the new direction added to the basis is the projected full gradient

$$g_{\text{basis}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w(v). \quad (45)$$

Thus the basis is enlarged by the actual gradient direction of the combined score. The old sketch energy, current-window energy, and pooled row entropy are already combined inside this single full-gradient vector.

In a reduced-subspace version, suppose the current local right search basis is

$$Z \in \mathbb{R}^{d \times q}, \quad Z^\top Z = I. \quad (46)$$

After solving the score problem inside $\text{span}(Z)$, the relevant expansion residual is the part of the feasible tangent gradient that is not already represented by this local basis:

$$r_Z(v) := (I - ZZ^\top) (I - QQ^\top - vv^\top) \nabla \text{score}_w(v). \quad (47)$$

The local basis should be expanded only when this residual is non-negligible. A natural stopping rule is

$$\frac{\|r_Z(v)\|_2}{\|(I - QQ^\top - vv^\top) \nabla \text{score}_w(v)\|_2} \leq \tau, \quad (48)$$

where τ is a prescribed tolerance. If the ratio is larger than τ , the new basis direction should be the normalized residual $r_Z(v)/\|r_Z(v)\|_2$, optionally followed by Krylov refinements generated from this residual direction.

Operator extraction from the retained right subspace. After the scored directions and any auxiliary directions are assembled into an orthonormal right basis

$$V \in \mathbb{R}^{d \times r}, \quad (49)$$

with row-wise representation $V^\top = V_t$, form

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}, \quad Y := M_w V. \quad (50)$$

Orthonormalize the image:

$$U := \text{orth}(Y) = \text{orth}(M_w V). \quad (51)$$

Equivalently, this defines the orthogonal projector onto the left image of the retained right space,

$$P_U := UU^\top. \quad (52)$$

Compress M_w to this left subspace:

$$\widetilde{M} := U^\top M_w, \quad (53)$$

which is the coordinate form of $P_U M_w$. Then compute

$$\widetilde{M} = U_{\text{small}} \Sigma V_{\text{new}}^\top, \quad (54)$$

and store

$$U_{\text{proj}} = U U_{\text{small}}, \quad S = \Sigma, \quad V^\top = V_{\text{new}}^\top. \quad (55)$$

The carried singular values and right singular vectors are not merely the raw scored-plus-auxiliary basis; they are the singular factors of the left-projected operator $P_U M_w$ restricted to the retained right span.

First-window special case. If $B = 0$ and $R = \emptyset$, then

$$c_1 = \frac{\log(\text{row}(A_1)/n)}{2 \log \text{row}(A_1)}, \quad (56)$$

and

$$\text{score}_1(v) = \|A_1 v\|_2^2 \|A_1 v\|_4^{4c_1} \|A_1 v\|_2^{-4c_1}. \quad (57)$$

The full gradient reduces to

$$\nabla \text{score}_1(v) = \phi_1(v) 2A_1^\top A_1 v \quad (58)$$

$$+ \|A_1 v\|_2^2 \phi_1(v) \left[4c_1 A_1^\top \left(\frac{(A_1 v)^{\odot 3}}{\|A_1 v\|_4^4} \right) - 4c_1 \frac{A_1^\top A_1 v}{\|A_1 v\|_2^2} \right], \quad (59)$$

where

$$\phi_1(v) = \|A_1 v\|_4^{4c_1} \|A_1 v\|_2^{-4c_1}. \quad (60)$$

Practical remarks. The objective is nonconvex, so multiple restarts are useful. One should reject or regularize steps when $\|C_w v\|_2$, $\|C_w v\|_4$, or $\|Bv\|_2^2 + \|A_w v\|_2^2$ is numerically tiny in a formula that divides by that quantity. The expanded gradient form avoids forming C_w explicitly, but the pooled norms must still include both Rv and $A_w v$.

The row sample should not be interpreted as a replacement for the old sketch. The sketch B supplies retained old energy, while R supplies raw old rows for the pooled entropy. A pure reservoir gives coverage, but it can miss rare large rows that dominate ℓ_4 terms. A practical row sample should therefore combine a uniform reservoir with a small set of rows having large entries on current sketch directions.